



## Chapter 5: Algebra

Three-week revision booklet • 6 worksheets • 240 questions

NSW syllabus outcome

MA4-ALG-C-01 — generalises number properties to operate with algebraic expressions • MAO-WM-01

NAME \_\_\_\_\_

### How to use this booklet

There are six worksheets, two for each week. Each one drills a single skill from the Algebra topic test. Each sheet has the same shape: **40 questions**, split into ● **Easy**, ● **Medium** and ● **Challenging**. Work from the top of the page down — the questions get harder as you go. Show your working for the challenging questions. The final worksheet mixes every skill together, so it is the best check that you are ready for the topic test.

| Week | Worksheet | Skill                                 |
|------|-----------|---------------------------------------|
| 1    | 1         | Laws of arithmetic                    |
| 1    | 2         | Writing algebraic expressions         |
| 2    | 3         | Simplifying and collecting like terms |
| 2    | 4         | Multiplying and dividing terms        |
| 3    | 5         | Substitution                          |
| 3    | 6         | Mixed revision — all skills           |

### Progress tracker

| Worksheet                               | Date completed | Score / 40 | Checked by |
|-----------------------------------------|----------------|------------|------------|
| 1 Laws of arithmetic                    |                |            |            |
| 2 Writing algebraic expressions         |                |            |            |
| 3 Simplifying and collecting like terms |                |            |            |
| 4 Multiplying and dividing terms        |                |            |            |
| 5 Substitution                          |                |            |            |
| 6 Mixed revision — all skills           |                |            |            |

# Worksheet 1: Laws of arithmetic

Week 1 · Worksheet 1 of 6 · 40 questions

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The distributive law lets you split a multiplication:  $8 \times 9 = 8 \times 10 - 8$ . The commutative and associative laws let you reorder and regroup to make a calculation easier.

## EASY

20 QUESTIONS

Use the distributive law to evaluate each expression.

- |                        |                          |                        |                         |
|------------------------|--------------------------|------------------------|-------------------------|
| 1. $21 \times 9$ _____ | 2. $18 \times 11$ _____  | 3. $34 \times 9$ _____ | 4. $27 \times 11$ _____ |
| 5. $16 \times 9$ _____ | 6. $45 \times 11$ _____  | 7. $23 \times 9$ _____ | 8. $52 \times 11$ _____ |
| 9. $19 \times 9$ _____ | 10. $38 \times 11$ _____ |                        |                         |

Use the commutative and associative laws to evaluate each expression.

- |                                   |                                   |                                   |                         |
|-----------------------------------|-----------------------------------|-----------------------------------|-------------------------|
| 11. $8 + 27 + 2$ _____            | 12. $4 \times 23 \times 25$ _____ | 13. $5 \times 17 \times 2$ _____  | 14. $36 + 19 + 4$ _____ |
| 15. $2 \times 49 \times 5$ _____  | 16. $7 + 58 + 3$ _____            | 17. $20 \times 17 \times 5$ _____ | 18. $6 + 45 + 4$ _____  |
| 19. $25 \times 13 \times 4$ _____ | 20. $50 \times 11 \times 2$ _____ |                                   |                         |

## MEDIUM

12 QUESTIONS

Use the distributive law to evaluate each expression.

- |                          |                          |                          |
|--------------------------|--------------------------|--------------------------|
| 21. $6 \times 43$ _____  | 22. $7 \times 52$ _____  | 23. $8 \times 31$ _____  |
| 24. $9 \times 24$ _____  | 25. $5 \times 68$ _____  | 26. $4 \times 87$ _____  |
| 27. $3 \times 96$ _____  | 28. $6 \times 74$ _____  | 29. $8 \times 49$ _____  |
| 30. $5 \times 198$ _____ | 31. $7 \times 101$ _____ | 32. $9 \times 102$ _____ |

## CHALLENGING

8 QUESTIONS

Answer each question, showing the law you used.

- |                                                                  |                                                                                   |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 33. Evaluate $12 \times 99$<br>_____                             | 34. Evaluate $15 \times 101$<br>_____                                             |
| 35. Evaluate $25 \times 36$<br>_____                             | 36. Evaluate $16 \times 125$<br>_____                                             |
| 37. Explain how to work out $8 \times 97$ in your head.<br>_____ | 38. Show that $4 \times 23 \times 25 = 2300$ .<br>_____                           |
| 39. Is $12 - 5$ the same as $5 - 12$ ? Explain briefly.<br>_____ | 40. Rewrite $7 \times 26$ as $7 \times 20 + 7 \times 6$ and evaluate it.<br>_____ |

## Worksheet 2: Writing expressions

Week 1 • Worksheet 2 of 6 • 40 questions

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In each question, use  $n$  to stand for the number.

### EASY

20 QUESTIONS

Write an algebraic expression for each statement. Use  $n$ .

- |                                     |                                                |
|-------------------------------------|------------------------------------------------|
| 1. 5 times the number _____         | 2. the number decreased by 7 _____             |
| 3. one-quarter of the number _____  | 4. the next consecutive number after $n$ _____ |
| 5. the number increased by 12 _____ | 6. 9 less than the number _____                |
| 7. double the number _____          | 8. the number divided by 3 _____               |
| 9. the number plus 4 _____          | 10. half of the number _____                   |
| 11. 8 more than the number _____    | 12. the number multiplied by 6 _____           |
| 13. the number take away 15 _____   | 14. three times the number _____               |
| 15. one-fifth of the number _____   | 16. the number less 1 _____                    |
| 17. 20 more than the number _____   | 18. the number shared between 8 _____          |
| 19. seven times the number _____    | 20. the number added to 30 _____               |

### MEDIUM

12 QUESTIONS

Write an algebraic expression for each statement. Use  $n$ .

- |                                                      |                                                   |
|------------------------------------------------------|---------------------------------------------------|
| 21. 3 more than twice the number _____               | 22. 5 less than three times the number _____      |
| 23. the sum of the number and 8, divided by 2 _____  | 24. four times the number, then take away 9 _____ |
| 25. the number squared _____                         | 26. 10 less than half the number _____            |
| 27. the number increased by 6, then doubled _____    | 28. seven times the number, plus 2 _____          |
| 29. the number divided by 4, then add 5 _____        | 30. one more than the number, times 3 _____       |
| 31. the previous consecutive number before $n$ _____ | 32. the number subtracted from 50 _____           |

### CHALLENGING

8 QUESTIONS

Write an expression for each situation.

- |                                                                                      |                                                                                             |
|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 33. Ana has $n$ marbles. Ben has 6 more. How many does Ben have?<br>_____            | 34. A pen costs \$ $n$ . What do 5 pens cost?<br>_____                                      |
| 35. A rectangle is $n$ cm long and 4 cm wide. Write its perimeter.<br>_____          | 36. Jo is $n$ years old. How old will she be in 12 years?<br>_____                          |
| 37. A pizza is cut into 8 slices. Write the fraction $n$ slices represents.<br>_____ | 38. Tickets cost \$12 each. Write the cost of $n$ tickets.<br>_____                         |
| 39. A number is doubled and then 7 is added. Write the expression.<br>_____          | 40. Write an expression for the sum of three consecutive numbers starting at $n$ .<br>_____ |

# Worksheet 3: Simplifying and like terms

Week 2 · Worksheet 3 of 6 · 40 questions

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Like terms have exactly the same pronumerals. Only like terms can be added or subtracted.

## EASY

20 QUESTIONS

Simplify each expression.

- |                         |                                |                                         |                                 |
|-------------------------|--------------------------------|-----------------------------------------|---------------------------------|
| 1. $x + x + x$ _____    | 2. $3 \times a \times 4$ _____ | 3. $7 \times p \times q \times 2$ _____ | 4. $5m + 3m$ _____              |
| 5. $9y - 4y$ _____      | 6. $2 \times b \times b$ _____ | 7. $6c + c$ _____                       | 8. $8k - 8k$ _____              |
| 9. $4 \times 5t$ _____  | 10. $d + d + d + d$ _____      | 11. $11w - 3w$ _____                    | 12. $5 \times e \times 3$ _____ |
| 13. $7g + 2g + g$ _____ | 14. $10h - 6h$ _____           | 15. $2 \times m \times n$ _____         | 16. $9r - 5r$ _____             |
| 17. $3v + 8v$ _____     | 18. $6 \times 2s$ _____        | 19. $12z - 7z$ _____                    | 20. $f + 4f$ _____              |

## MEDIUM

12 QUESTIONS

Simplify each expression by collecting like terms.

- |                               |                                   |                               |
|-------------------------------|-----------------------------------|-------------------------------|
| 21. $9xy + 4 + 6xy$ _____     | 22. $8h^2 + 3h - 2h^2 + 5h$ _____ | 23. $5a + 3b + 2a + 4b$ _____ |
| 24. $12p - 5q - 4p$ _____     | 25. $7m + 2 - 3m + 8$ _____       | 26. $6de + 2d - de$ _____     |
| 27. $4x^2 + 9x - x^2$ _____   | 28. $10k + 3j - 6k - j$ _____     | 29. $2ab + 5ba$ _____         |
| 30. $15t - 4 - 9t + 11$ _____ | 31. $3c^2 + 7c + c^2 - 2c$ _____  | 32. $8mn - 3nm + 4m$ _____    |

## CHALLENGING

8 QUESTIONS

Simplify each expression, or answer the question.

- |                                                                                                                |                                                                     |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 33. $6a + 4b - 2a - 9b + 5a$<br>_____                                                                          | 34. $3x^2 + 5x - x^2 - 8x + 2$<br>_____                             |
| 35. $7pq + 3p - 2qp + 5q - p$<br>_____                                                                         | 36. $2m + 3n - 4m + 6 - 2n - 1$<br>_____                            |
| 37. Why can $5ab$ and $3ba$ be added, but $5ab$ and $3a$ cannot?<br>_____                                      | 38. A student wrote $4x + 3y = 7xy$ . Explain the mistake.<br>_____ |
| 39. Write an expression for the perimeter of a rectangle with sides $3a$ and $2b$ , then simplify it.<br>_____ | 40. Simplify $5k \times 2 + 3k - k$<br>_____                        |

# Worksheet 4: Multiplying and dividing terms

Week 2 • Worksheet 4 of 6 • 40 questions

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Multiply the numbers first, then the pronumerals. When dividing, cancel common factors.

## EASY

20 QUESTIONS

Simplify each expression.

- |                         |                          |                          |                          |
|-------------------------|--------------------------|--------------------------|--------------------------|
| 1. $4m \times 3m$ _____ | 2. $6v \times 5$ _____   | 3. $20m \div 5$ _____    | 4. $18xy \div 6x$ _____  |
| 5. $2a \times 7$ _____  | 6. $5b \times 4b$ _____  | 7. $24k \div 8$ _____    | 8. $9p \times 2q$ _____  |
| 9. $30t \div 6$ _____   | 10. $3c \times 3c$ _____ | 11. $16ab \div 4a$ _____ | 12. $8 \times 7n$ _____  |
| 13. $45y \div 9$ _____  | 14. $2x \times 6y$ _____ | 15. $12mn \div 3n$ _____ | 16. $7d \times d$ _____  |
| 17. $36w \div 12$ _____ | 18. $4e \times 5f$ _____ | 19. $21pq \div 7q$ _____ | 20. $10g \times 3$ _____ |

## MEDIUM

12 QUESTIONS

Simplify each expression.

- |                                   |                                    |                                   |
|-----------------------------------|------------------------------------|-----------------------------------|
| 21. $2a \times 3b \times 4$ _____ | 22. $40x^2 \div 8x$ _____          | 23. $5m \times 4m \times 2$ _____ |
| 24. $56abc \div 7ab$ _____        | 25. $6p \times 2p \times 3p$ _____ | 26. $72y^2 \div 9y$ _____         |
| 27. $3k \times 5k \times k$ _____ | 28. $48mn \div 6mn$ _____          | 29. $9r \times 2s \times 2$ _____ |
| 30. $100c^2 \div 25c$ _____       | 31. $4d \times 3d \times 5$ _____  | 32. $63xyz \div 9xz$ _____        |

## CHALLENGING

8 QUESTIONS

Simplify each expression, or answer the question.

- |                                                                                 |                                                                    |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------|
| 33. $12a \times 3b \div 4ab$<br>_____                                           | 34. $(5m \times 4m) \div 10m$<br>_____                             |
| 35. $8x^2y \div 2xy$<br>_____                                                   | 36. $6p \times 2q \times 3pq$<br>_____                             |
| 37. A rectangle has sides $4a$ and $6a$ . Write and simplify its area.<br>_____ | 38. A square has side $5k$ . Write and simplify its area.<br>_____ |
| 39. Explain why $3m \times 3m$ is $9m^2$ and not $6m^2$ .<br>_____              | 40. Simplify $(20ab \div 5b) \times 2a$<br>_____                   |

# Worksheet 5: Substitution

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Replace each pronumeral with its value, then work out the answer using the order of operations. Throughout this sheet, **a = 4, b = 3, c = 5**.

## EASY

20 QUESTIONS

If **a = 4, b = 3** and **c = 5**, evaluate each expression.

- |                      |                   |                    |                    |
|----------------------|-------------------|--------------------|--------------------|
| 1. $a + b + c$ _____ | 2. $2a - c$ _____ | 3. $a + b$ _____   | 4. $bc$ _____      |
| 5. $c - b$ _____     | 6. $3a$ _____     | 7. $abc$ _____     | 8. $a + 2b$ _____  |
| 9. $10 - c$ _____    | 10. $4b$ _____    | 11. $c - a$ _____  | 12. $2b + c$ _____ |
| 13. $ac$ _____       | 14. $a - b$ _____ | 15. $b + c$ _____  | 16. $5a$ _____     |
| 17. $ab$ _____       | 18. $c + 3$ _____ | 19. $2c - b$ _____ | 20. $a + c$ _____  |

## MEDIUM

12 QUESTIONS

If **a = 4, b = 3** and **c = 5**, evaluate each expression.

- |                        |                          |                            |
|------------------------|--------------------------|----------------------------|
| 21. $a^2$ _____        | 22. $b^2 + c$ _____      | 23. $2a + 3b - c$ _____    |
| 24. $ab - c$ _____     | 25. $c^2 - a$ _____      | 26. $3(a + b)$ _____       |
| 27. $abc \div 2$ _____ | 28. $5c - 2a$ _____      | 29. $\frac{ac}{b+1}$ _____ |
| 30. $a^2 - b^2$ _____  | 31. $2(b + c) - a$ _____ | 32. $4a - b - c$ _____     |

## CHALLENGING

8 QUESTIONS

Evaluate, or answer the question. Use **a = 4, b = 3, c = 5**.

- |                                                                                                 |                                                                                    |
|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 33. $a^2 + b^2 + c^2$<br>_____                                                                  | 34. $(a + b)^2 - c$<br>_____                                                       |
| 35. $2a^2 - 3b$<br>_____                                                                        | 36. $\frac{abc}{a+b+c}$<br>_____                                                   |
| 37. Which is larger, $a^2b$ or $ab^2$ ? Show your working.<br>_____                             | 38. If $P = 2l + 2w$ , find $P$ when $l = 4$ and $w = 3$ .<br>_____                |
| 39. If $A = \frac{1}{2}bh$ , find $A$ when $b = 8$ and $c$ is not needed and $h = 5$ .<br>_____ | 40. Find the value of $3a - 2b + c$ when $a = -4$ , $b = 3$ and $c = 5$ .<br>_____ |

# Worksheet 6: Mixed revision

Week 3 • Worksheet 6 of 6 • 40 questions

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This sheet mixes every skill from the Algebra chapter. Work through it as though it were the topic test.

## EASY

20 QUESTIONS

Simplify each expression.

- |                          |                                |                                  |                          |
|--------------------------|--------------------------------|----------------------------------|--------------------------|
| 1. $y + y + y + y$ _____ | 2. $5 \times k \times 2$ _____ | 3. $8p + 3p$ _____               | 4. $14m - 6m$ _____      |
| 5. $3a \times 4a$ _____  | 6. $24x \div 6$ _____          | 7. $2c \times 5d$ _____          | 8. $20mn \div 4n$ _____  |
| 9. $7t - 7t$ _____       | 10. $6 \times 3w$ _____        | 11. $9h + 2h - h$ _____          | 12. $15y \div 5$ _____   |
| 13. $4e \times e$ _____  | 14. $11q - 4q$ _____           | 15. $2 \times r \times 3s$ _____ | 16. $18ab \div 9b$ _____ |

Use a law of arithmetic to evaluate each expression.

- |                         |                          |                                  |                        |
|-------------------------|--------------------------|----------------------------------|------------------------|
| 17. $25 \times 9$ _____ | 18. $16 \times 11$ _____ | 19. $5 \times 27 \times 2$ _____ | 20. $9 + 46 + 1$ _____ |
|-------------------------|--------------------------|----------------------------------|------------------------|

## MEDIUM

12 QUESTIONS

Simplify or evaluate each expression.

- |                                                |                                                |                                                 |
|------------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 21. $6xy + 2 + 3xy$ _____                      | 22. $5a^2 + 2a - a^2$ _____                    | 23. $7m + 4n - 2m - n$ _____                    |
| 24. $30p^2 \div 6p$ _____                      | 25. $3k \times 2k \times 4$ _____              | 26. $42abc \div 6ac$ _____                      |
| 27. If $a = 4$ , $b = 3$ : find $3a - b$ _____ | 28. If $a = 4$ , $b = 3$ : find $ab + 5$ _____ | 29. If $a = 4$ , $b = 3$ : find $a^2 - b$ _____ |
| 30. Write: 6 less than twice $n$ _____         | 31. Write: one-third of $n$ _____              | 32. Write: the number after $n$ _____           |

## CHALLENGING

8 QUESTIONS

Answer each question. Show your working.

- |                                                                                                     |                                                                                                      |
|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 33. Simplify $8a + 5b - 3a - 7b + 2a$<br>.....                                                      | 34. Simplify $(24x^2y \div 6xy) \times 3x$<br>.....                                                  |
| 35. If $a = 4$ , $b = 3$ and $c = 5$ , evaluate $2a^2 - bc$<br>.....                                | 36. Evaluate $14 \times 99$ using the distributive law.<br>.....                                     |
| 37. A rectangle has length 5m and width 3m. Write and simplify its perimeter and its area.<br>..... | 38. A student wrote $6p \times 2p = 12p$ . Explain the mistake and give the correct answer.<br>..... |
| 39. Write an expression for the total cost of $n$ books at \$9 each plus \$4 postage.<br>.....      | 40. Simplify $4c^2 + 3c - c^2 + 8 - 5c$<br>.....                                                     |